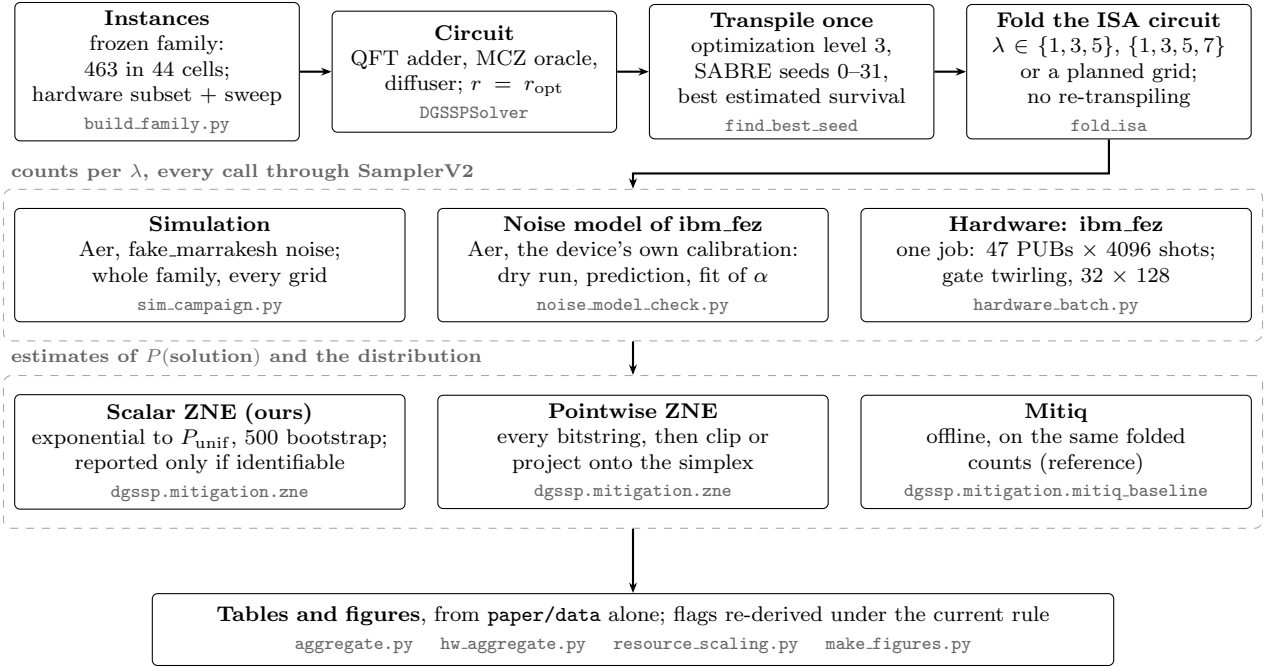

Instances
frozen family:
463 in 44 cells;
hardware subset + sweep
build_family.py
Circuit
QFT adder, MCZ oracle,
diffuser; $r = r_{\text{opt}}$
DGSSPSolver
Transpile once
optimization level 3,
SABRE seeds 0–31,
best estimated survival
find_best_seed
Fold the ISA circuit
$\lambda \in \{1, 3, 5\}$, $\{1, 3, 5, 7\}$
or a planned grid;
no re-transpiling
fold_isa
counts per λ, every call through SamplerV2
Simulation
Aer, fake_marrakesh noise;
whole family, every grid
sim_campaign.py
Noise model of ibm_fez
Aer, the device's own calibration:
dry run, prediction, fit of α
noise_model_check.py
Hardware: ibm_fez
one job: 47 PUBs × 4096 shots;
gate twirling, 32 × 128
hardware_batch.py
estimates of P(solution) and the distribution
Scalar ZNE (ours)
exponential to P_{unif}, 500 bootstrap;
reported only if identifiable
dgssp.mitigation.zne
Pointwise ZNE
every bitstring, then clip or
project onto the simplex
dgssp.mitigation.zne
Mitiq
offline, on the same folded
counts (reference)
dgssp.mitigation.mitiq_baseline
Tables and figures, from paper/data alone; flags re-derived under the current rule
aggregate.py
hw_aggregate.py
resource_scaling.py
make_figures.py